

IMPORTANT TOPICS FOR FINAL

Oscillation

1. Differential eq of SHM
1. graphical representation of SHM
2. Solution of Simple harmonic motion
3. Total energy of SHM
4. average kinetic energy
5. mathematical problem

Waves

1. standing wave (with graph, antinode, node)
2. dopler effect (4 types)
- [3. beats](#)
4. dopler and beats related math problem

Thermodynamics

1. Carno cycle
2. why C_p is greater than C_v
3. Thermodynamics law
4. math problem

Optics

1. Interference of two light waves: analytical treatment
2. Distinction between Fresnel and Fraunhofer diffraction:
3. Intensity pattern due to single slit diffraction The maxima and minima of the pattern. (graph)
4. Brewster's law (Statement and explanation)
5. young double slit
6. math related problem

All Definition